ELEN 3803 Electronic Circuits Lab 2 Diode

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February 2026

1 Experiment 1: Measurement of Diode I-V Characteristics

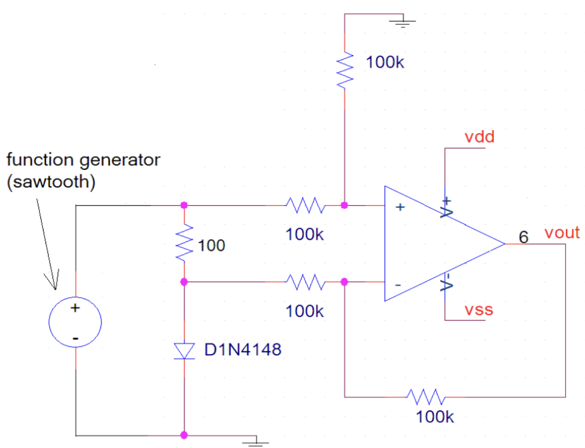


Figure 1: Source: Lab 2 Manual

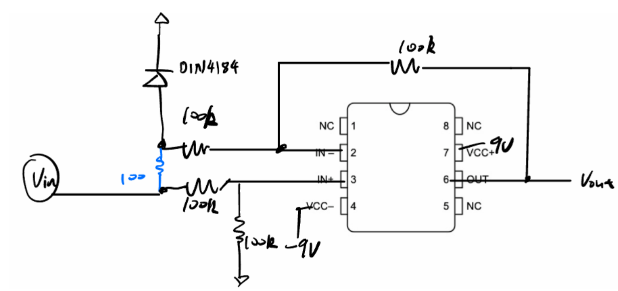


Figure 2: Layout

Curve-tracer front-end, with diode under test.

Current Conversion Factor: The vertical scale conversion from voltage to current is determined by the 100 Ω sense resistor. The conversion is:

$$I_D = \frac{V_R}{100\,\Omega} = 1.25V/100\Omega = 0.125 \text{ mA}$$

Figure 3: 1N4148 I-V curve. Division 1 & 1.25 V

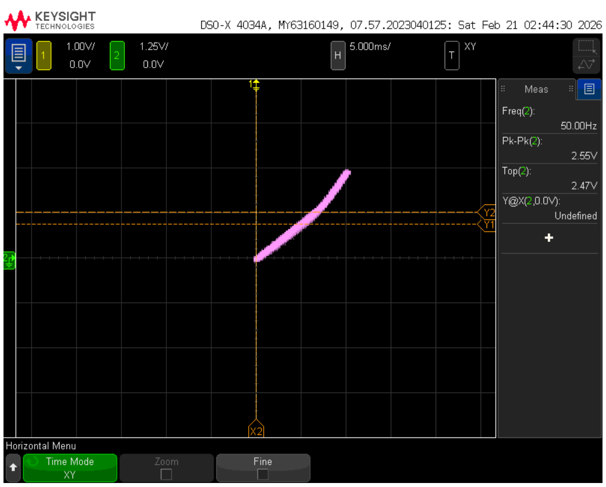


Figure 4: Red LED I–V Curve

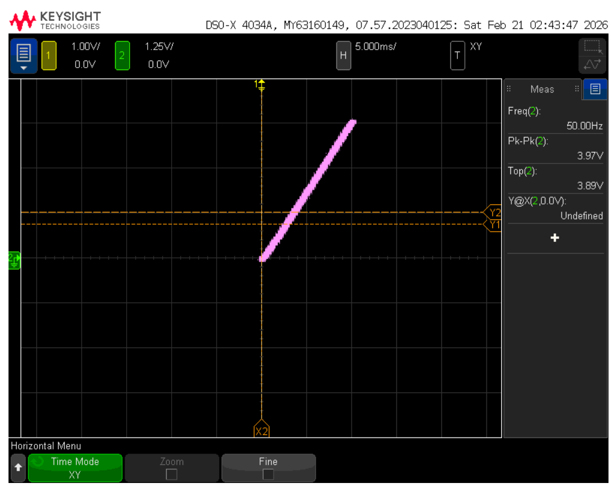


Figure 5: Germanium Schottky I–V Curve

2 Experiment 2: MXR Distortion Plus Guitar Pedal

Above we have the calculations for the voltages. Below is the measured DC voltage, 4.78 Volts, which doesn't contradict our calculation.

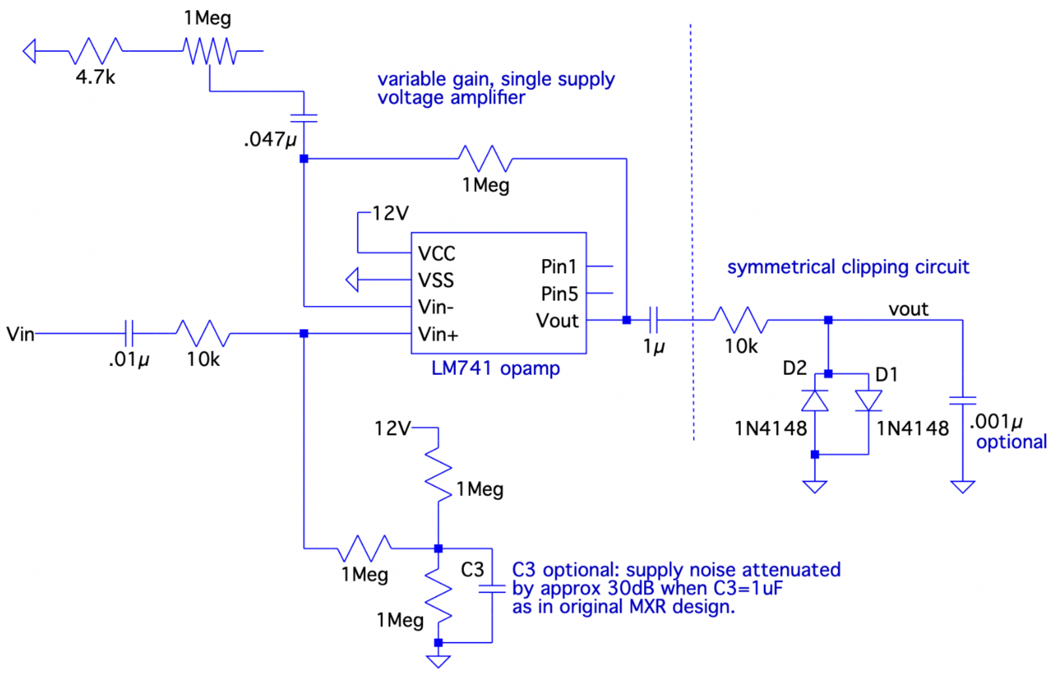


Figure 6: Source: Lab 2 Manual

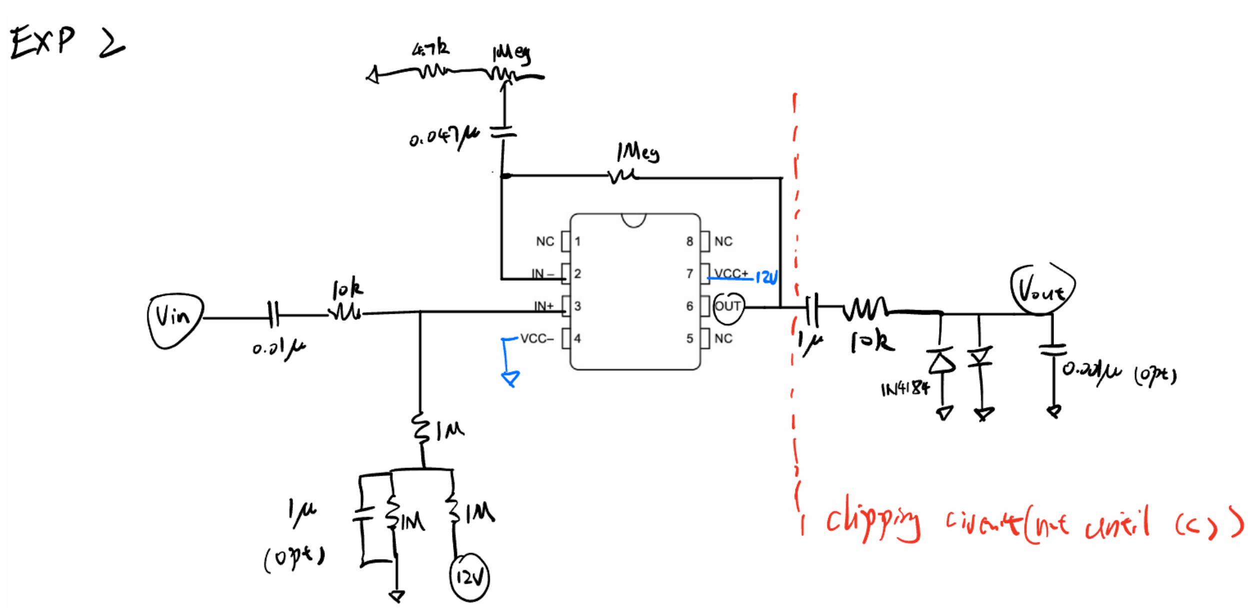
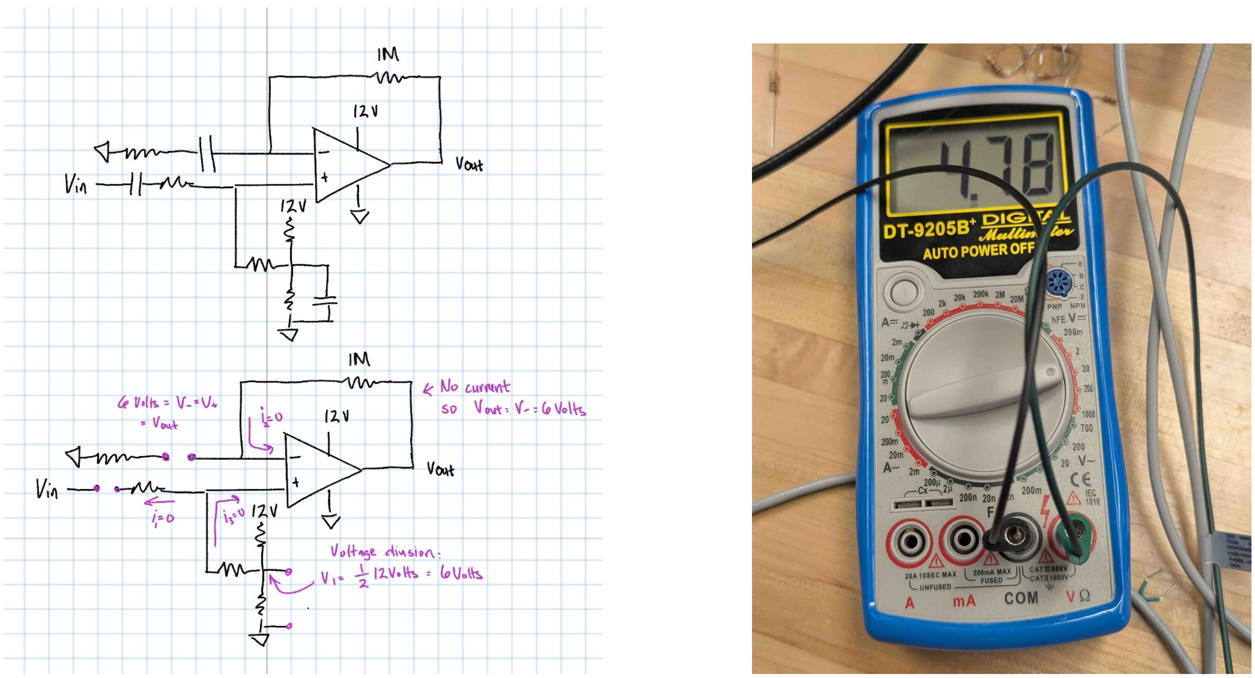


Figure 7: Layout

2.1 Part (a)

- Calculated V_{in+}: 6V
- Calculated V_{in-}: 6V
- Calculated V_{out}: 6V
- These DC voltages are used to bias the op-amp in a single-supply configuration, typically at half the supply voltage to allow for maximum signal swing.



(a) DC Calculation (b) Physical Measurement

Figure 8

2.2 Part (b)

The gain is adjusted using a $1\,M\Omega$ potentiometer.

Following that we have a non-inverting amplifier configuration and 1KHz input,
Our gain is $1 + R_2/R_1$.
With R2 = 1 M Ω and R1 varying from 1M Ω + 4.7k to 4.7k,

- **Minimum Gain (Pot = 1 MΩ):** 2
- **Maximum Gain (Pot $\approx$ 0 Ω):** 200

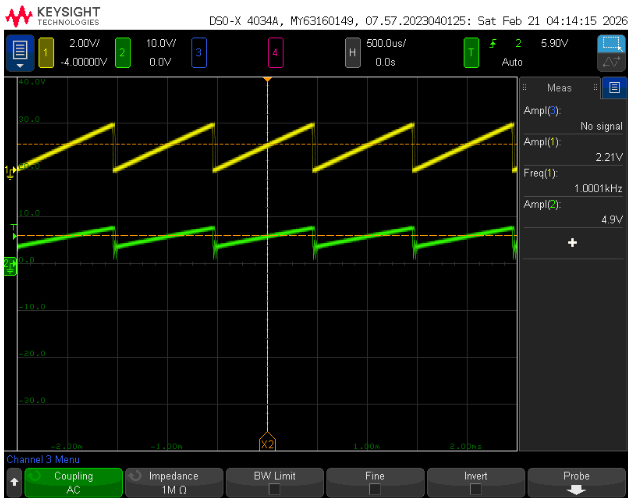


Figure 9: Exp2 Part (b): Minimum gain is ≈ 2, as expected

2.3 Part (c) Clipping circuit

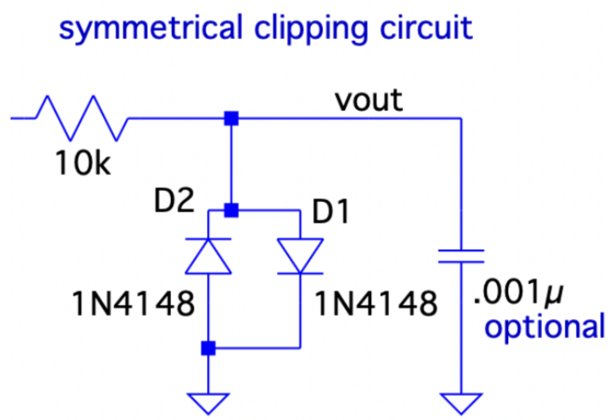


Figure 10: Clipping circuit schematic. Source: Lab Manual

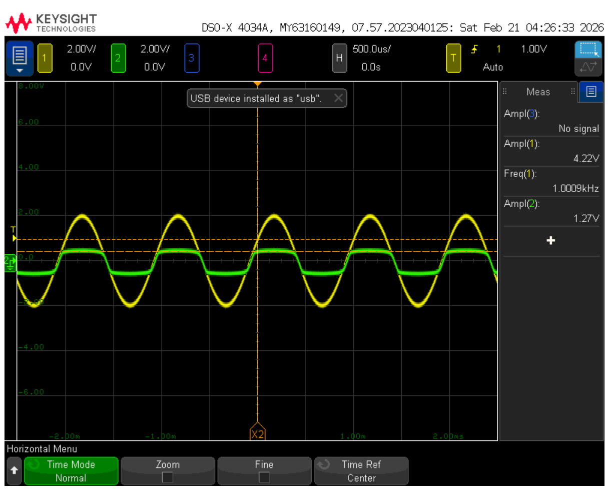
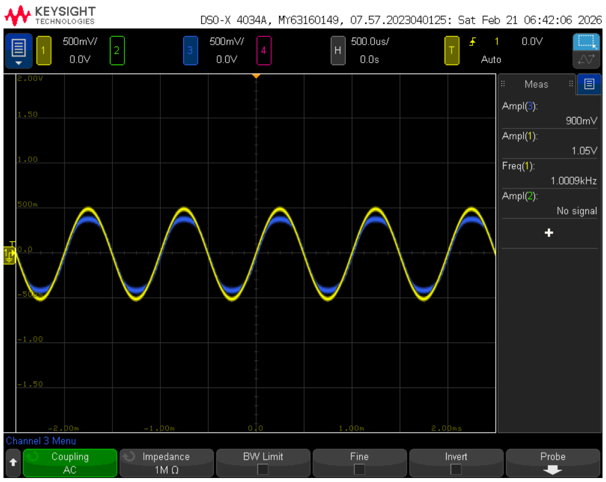
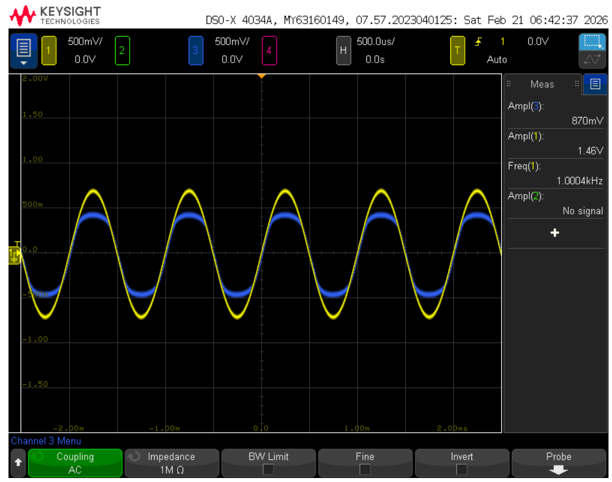


Figure 11: Part (c) Clipping Circuit — Normal Mode

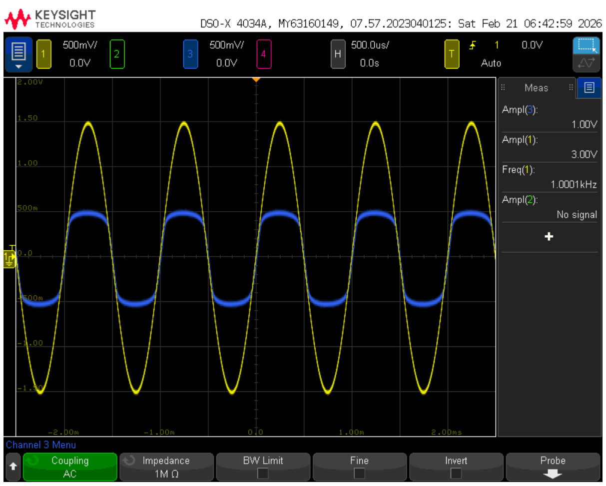
Figure 12: Part (c) Clipping Circuit — XY Mode



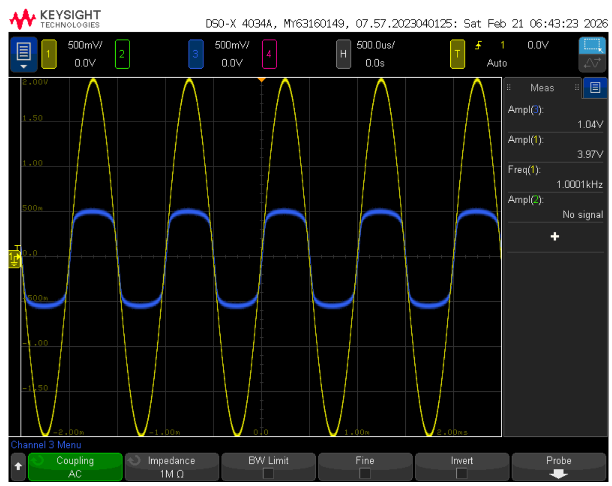
(a) Input = 1.05 Vpp



(b) Input = 1.46 Vpp



(c) Input = 3.00 Vpp



(d) Input = 4.00 Vpp

Figure 13: Output waveform clipping behavior at different input amplitudes. Yellow = input, Blue = output. The output did not significantly change when input is higher than 3.0Vpp

2.4 Part (d) Connecting clipper with Op Amp

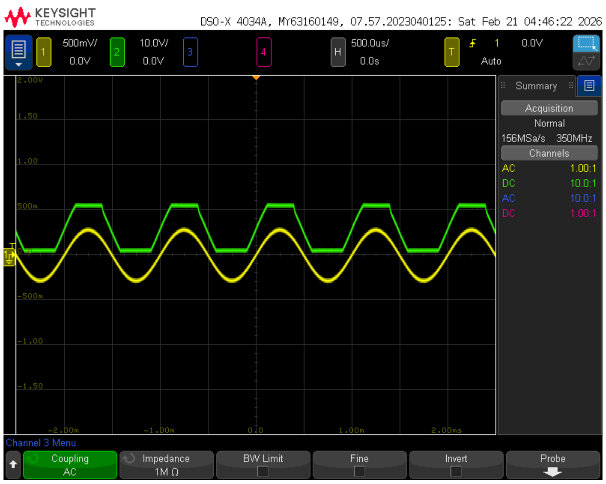


Figure 14: Part (d) OP Amp and Clipper — Case 1

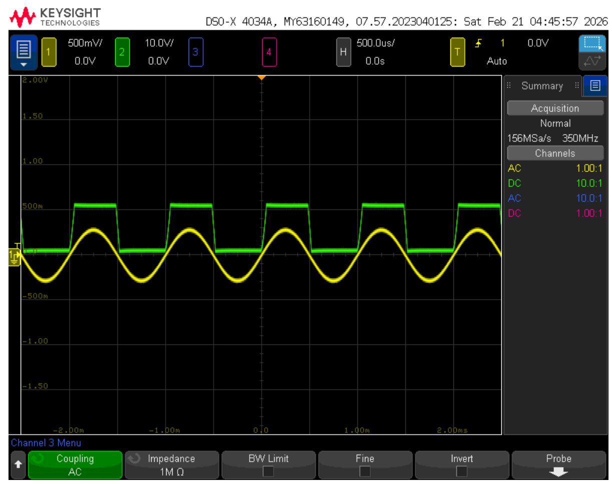


Figure 15: Part (d) OP Amp and Clipper — Case 2

The purpose of the variable amplifier being installed before the clipper is so that the amount of clipping and distortion can be varied. It's through the adjusting of the potentiometer's resistance that we were able to see how clipped or not our output to the guitar waveform was.

At low gain, the signal stays below 0.7V, and the sound is clean. At high gain, the amplifier forced the voltage to 4V or higher. Since the diodes won't let it pass 0.7V, the signal is flatten, creating intense distortion.

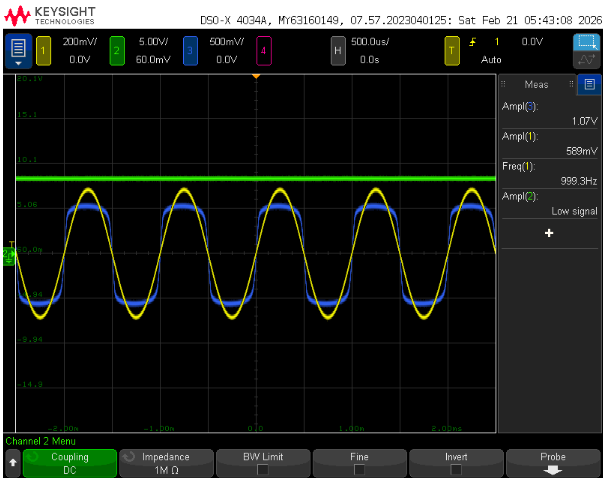
2.5 Part(e) Audio

Figure 16: Part(e) Audio signal

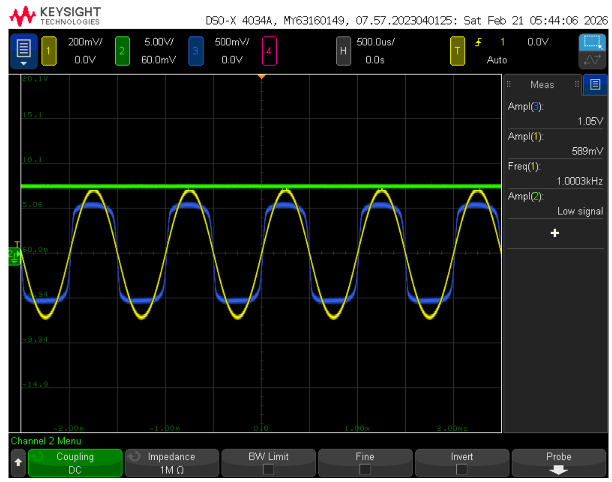
3 Challenge

Design was approved by the LA.

3.1 Effect of Supply Voltage Change



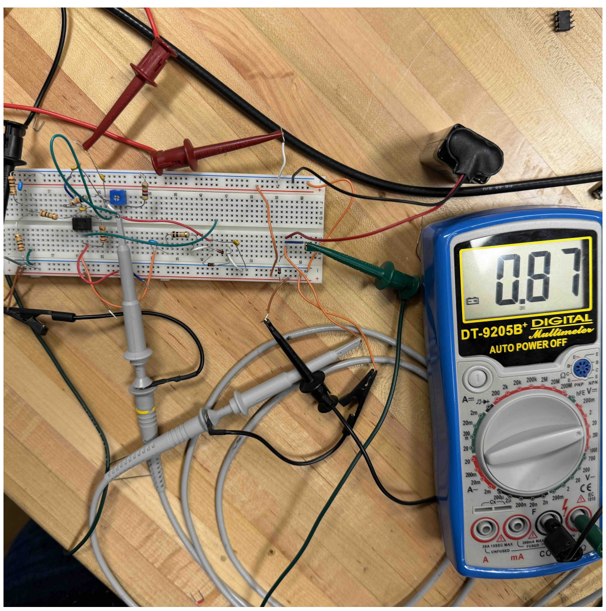
(a) 12V and 9V both on



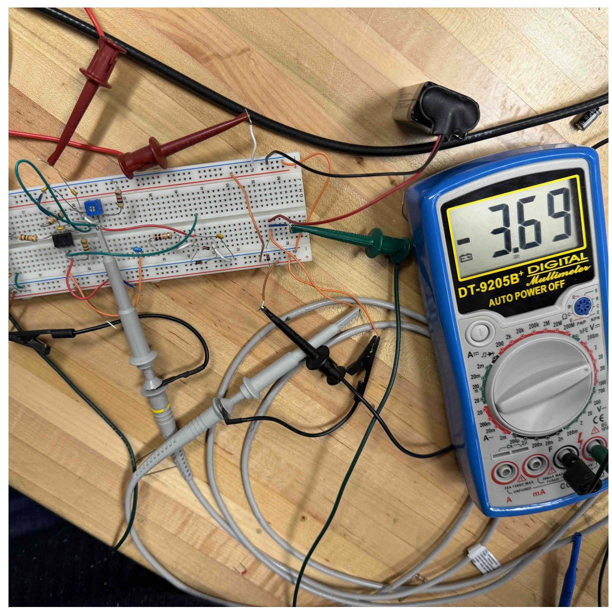
(b) 12V removed, 9V only

Figure 17: Negligible difference

3.2 Diode verification



(a) Diode is forward biased, turn on voltage is about 0.7V



(b) Diode is reversed biased when both 12V and 9V are on.

Figure 18: Both case indicate the diode is operating properly

When both 12V and 9V on, the voltage across the diode is around 3V. When only 9V on, the voltage across the diode is around 0.7V. Both case indicate the diode is operating properly.